



**ENVIRONMENT
AGENCY**

Performance Standards for Continuous Ambient Air Quality Monitoring Systems

**Environment Agency
September 2005
Version 4**



Foreword

The Environment Agency has established its Monitoring Certification Scheme MCERTS, to deliver quality environmental measurements. The Scheme is based on international standards and provides for the product certification of instruments, the competency certification of personnel and the accreditation of laboratories. This document sets out the MCERTS performance standards for continuous ambient air quality monitoring systems (CAMs).

The benefits of MCERTS are that it:

- makes available a certification scheme that is formally recognised within the UK and is accepted internationally;
- gives confidence to regulatory authorities that instrumentation, once certified, is fit for purpose and capable of producing results of the required quality and reliability;
- gives confidence to users of monitoring equipment that the instrumentation selected is robust and conforms to performance standards that are accepted by UK regulatory authorities;
- supports the supply of accurate and reliable data to the public;
- provides instrument manufacturing companies with an independent authoritative endorsement of their products, which will facilitate their access to international markets and increase the take-up of their products in the UK.

The MCERTS instrument performance standards described in this document are based on relevant sections of a number of international ISO and CEN standards. Close links have also been established with the CEN technical committee working groups currently developing new standards on ambient air pollution monitoring. The pollutants covered include nitrogen monoxide (NO), nitrogen dioxide (NO₂), sulphur dioxide (SO₂), carbon monoxide (CO), ozone (O₃), particulate matter (PM₁₀ and PM_{2.5}), lead, other metals (cadmium, arsenic, nickel and mercury), polycyclic aromatic hydrocarbons (PAHs) and benzene. As MCERTS is developed, the performance standards will be expanded to cover other pollutants.

MCERTS for continuous ambient air quality monitoring systems is a formal product certification scheme operating under the requirements of European Standard EN 45011. We have appointed Sira Certification Service (SCS) as the Certification Body to operate MCERTS on our behalf.

Laboratory and field testing under MCERTS must be carried out by laboratories and test organisations that are accredited to BS/EN ISO/IEC 17025. These organisations must also be capable of meeting the MCERTS performance standards for testing, as specified in a separate document under the Scheme. The evaluation of the results obtained during the laboratory and field tests is carried out by SCS, using a group of independent experts known as the Certification Committee.

If you have any questions about the certification process, or would like further information on how to make an application, please contact:

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For more information on MCERTS and for copies of the performance standards and further guidance, see the Environment Agency's website at:
www.mcerts.net

Record of amendments

Version number	Date	Amendment
Version 4	July 2005	Alignment with the requirements of the Ambient Air Quality Standards: BS EN 14212:2005, BS EN14625:2005 BS EN 14211:2005, BS EN 14625:2005
Version 4	July 2005	Introduction of the methodology introduced by the EC Working Group on Guidance for Demonstration of Equivalence of Ambient Air Monitoring Methods into ambient dust monitoring equipment test requirements. Removal of wind tunnel and artificial dust generation based testing methods.
Version 4	August 2005	Modification of field repeatability calculation criteria for lower concentration scenarios reflecting reducing levels of ambient pollution for some species. Introduction of consistent criterion of unchanging test stringency.
Version 4	August 2005	Allowing for performance testing with the Kalman filter always switched on for single chamber nitrogen oxides monitors if they are intended for use in this mode of operation only.

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1 Introduction

- 1.1 This document describes the MCERTS Performance Standards for certain categories of continuous ambient air quality monitoring systems (CAMs). There is also a separate but related document describing the procedures for organisations involved in the testing of CAMS⁽¹⁾.
- 1.2 For the purpose of this document, CAMs are instrumental systems that either:
- continuously monitor ambient pollutant concentrations in situ and automatically produce results; or
 - sample ambient air over an extended period (eg days or weeks) onto a filter or a similar device, which is subsequently analysed in the laboratory to determine, for example, the amount of particulates, metals, or PAHs present. Such samplers may be used to obtain a continuous, or a discontinuous, longer-term average value of the pollutant concentration, by carrying out sequential sampling over time. For these instruments, the performance standards cover only the sampling stage of the measurement process.

Open-path instruments are covered by separate MCERTS specifications.

- 1.3 The following air pollutants are covered: nitrogen monoxide (NO), nitrogen dioxide (NO₂), sulphur dioxide (SO₂), carbon monoxide (CO), ozone (O₃), particulate matter (PM₁₀ and PM_{2.5}), lead, other metals (cadmium, arsenic, nickel and mercury), polycyclic aromatic hydrocarbons (PAHs) and benzene.
- 1.4 The main instrument performance characteristics against which a CAM will be assessed by a combination of laboratory and field testing are:
- response time, consisting of rise lag time, rise time, fall lag time and fall time (where applicable);
 - laboratory repeatability standard deviation;
 - zero and span drifts;
 - accuracy of sample collection in the case of certain particulate CAMs, and in the case of all CAMs that monitor metals and polycyclic aromatic hydrocarbons (PAHs)];
 - detection limit;
 - averaging of short-term fluctuations in determinand concentration (where applicable);
 - linear fit;
 - cross-sensitivity to interfering substances;
 - NO_x converter efficiency test (where applicable);
 - carry-over (where applicable);
 - flow accuracy and stability (where applicable);
 - influence of atmospheric sample pressure and temperature;
 - susceptibility to physical disturbances (where required);

- field performance of the CAM against a reference method where this is available, and/or against another CAM;
- field repeatability;
- long-term zero and span drift;
- availability (maintenance interval).

The definitions of the performance characteristics (and other terms used in this document) are given in Section 4.

- 1.5 The general requirements and the performance standards to be met by CAMS are presented in this document in Sections 5 and 6 respectively. The main published and draft international and national standards on which these performance standards are based are presented in Appendix 1. Appendix 2 lists the main PAHs of interest in air quality monitoring. The range of standard environmental operating conditions for CAMs is described in Appendix 3.
- 1.6 The MCERTS Performance Standards have been defined so that MCERTS-certified CAMs would be capable of meeting the requirements of both the EC Framework Directive 96/62 (EC “Ambient Air Quality Assessment and Management”)⁽²⁾ and its Associated Air Quality Daughter Directives. The total allowable uncertainties at the specified limit values for CAMs monitoring sulphur dioxide, oxides of nitrogen and PM₁₀ particulates are now published in the EC Air Quality Daughter Directive⁽³⁾. In addition, a Daughter Directive has been prepared to cover benzene and carbon monoxide and this also specifies the total allowable uncertainties at the proposed limit values for these pollutants⁽⁴⁾. The total allowable uncertainties at the specified limit values for arsenic, cadmium, nickel and polycyclic aromatic hydrocarbons are published in the EC Air Quality Daughter Directive⁽⁸⁾. The Ambient Air Quality CEN Standards define the technical requirements for monitoring of the pollutants covered by the Directives. A list of the relevant CEN Technical Committees and their associated Working Groups is provided in Section A1.2. The central objective in defining the MCERTS Performance Standards is that they are fully compatible with the CEN standards.
- 1.7 Throughout this document the terms “MCERTS certificate” and “certificate” refer to the MCERTS product-conformity certificate.

2 European Union and United Kingdom legislative requirements for continuous ambient air quality monitoring certification

2.1 Acceptance for use

- 2.1.1 The EC Ambient Air Quality Directives require that CAMs used to report air quality results undergo product certification or type approval before they are accepted for use. The MCERTS scheme is designed to meet these requirements and is recognised by the Department for Environment, Food and Rural Affairs (DEFRA) as a vehicle to deliver such product certification.

2.2 Air quality limit values

- 2.2.1 The performance standards specified in this document are generally expressed as measurement results from which the values of standard uncertainties can be derived. Tables 2.1 and 2.2 give the standards from the Air Quality Strategy for England, Scotland, Wales and Northern Ireland and the European Union air-quality limit values in absolute concentration units. The proposed EU limit values are, in general, very similar to the concentrations given in the UK's National Air Quality Standards NAQS⁽⁵⁾. The limit values that have been used to derive the performance characteristics specified in this document are given in Table 2.3. EU limit values have been chosen where these are available. Where these have not yet been defined, UK NAQS limit values have been adopted.
- 2.2.2 Where certification of CAMs is required in scenarios where no limit value is specified, 80% of the certification range shall be used as the value at which the performance characteristics are defined.
- 2.2.3 The UK currently requires 15 minute average values for SO₂ ambient air quality concentrations to be reported, whereas the current EU Directive specifies 10 minute averaging times. It is possible that the UK will propose a procedure that shows equivalence with the EU requirements but continue to report 15 minute SO₂ average concentrations. This will be taken into account by the relevant MCERTS Certification Committee if it arises.

Table 2.1 Air quality strategy for England, Scotland, Wales and Northern Ireland limit values

Pollutant	NAQS averaging time	NAQS standard	Number of permissible exceedances	Date to be met
SO ₂	15-min mean	100 ppb	max. of 35 exceedances	end 2005
	1-hour mean	132 ppb	max. of 24 exceedances	end 2004
	24-hour mean	47 ppb	max. of 3 exceedances	end 2004
	calendar year & winter	8 ppb	rural areas	2000
NO ₂	1-hour mean	105 ppb	max. of 18 exceedances	end 2005
	annual mean	21 ppb		end 2004
NO _x	annual mean	16 ppb		end 2000
PM ₁₀	running 24-hour mean	50 µg/m ³	max. of 35 exceedances	end 2004
	annual mean	40 µg/m ³	max. of 7 exceedances (under review)	end 2009
	annual mean	20 µg/m ³		
	24-hour mean	50 µg/m ³		
Pb	annual mean	0.5 µg/m ³ 0.25 µg/m ³	None	end 2004 end 2008
Benzene	running annual mean	5ppb 1 ppb	None	end 2003 end 2005
CO	running 8-hour mean	10 ppm 8.6 ppm	None	end 2003 end 2004
1,3, Butadiene	running annual mean	1 ppb	None	end 2003
O ₃	running 8-hour mean	50 ppb	max. of 10 exceedances	end 2005

Table 2.2 European Union air quality limit values

Pollutant	Period	Limit	Number of permissible exceedances	Date to be met
SO ₂	1 hour ^a	131 ppb	not to be exceeded > 24 times per calendar year	1 Jan 2005 ^b
	24 hours	48.6 ppb	not to be exceeded > 3 times per calendar year	1 Jan 2005
	calendar year & winter	7 ppb	rural areas	E.I.F. ^c + 24 months
NO ₂	1 hour	104.6 ppb	not to be exceeded > 18 times per calendar year	1 Jan 2010 ^b
NO _x	calendar year	21 ppb	rural areas	1 Jan 2010 ^b
	calendar year	15.7 ppb		E.I.F. + 24 months
PM ₁₀	24 hours	50 µg/m ³	not to be exceeded > 35 times per calendar year	1 Jan 2005 ^b
	calendar year	40 µg/m ³	not to be exceeded > 7 times per calendar year	1 Jan 2005
	24 hours	50 µg/m ³		1 Jan 2010
	calendar year	20 µg/m ³		1 Jan 2010 ^b
Pb	calendar year	0.5 µg/m ³	None	1 Jan 2005 ^d
Benzene	calendar year	1.66 ppb ^e	None	1 Jan 2010 ^b
CO	rolling 8-hour mean	8.6 ppm ^e	None	1 Jan 2005
O ₃	hourly mean	120 ppb	20 exceedances	1 Jan 2005
Arsenic	Annual average	6 ng/m ³	Total content in the PM ₁₀ fraction averaged over calendar year	31 Dec 2012
Cadmium	Annual average	5 ng/m ³	Total content in the PM ₁₀ fraction averaged over calendar year	31 Dec 2012
Nickel	Annual average	20 ng/m ³	Total content in the PM ₁₀ fraction averaged over calendar year	31 Dec 2012
Benzo(a)pyrene	Annual average	1 ng/m ³	Total content in the PM ₁₀ fraction averaged over calendar year	31 Dec 2012

^a Formed from 10-minute averages or 15-minute averages (UK)

^b A decreasing margin of tolerance is applicable until this date.

^c E.I.F. = entry into force (of EU Directive).

^d Or 2010 at industrial sites contaminated with lead

^e Currently proposed values

Table 2.3 Limit values used for MCERTS performance standards

Pollutant	Limit value
Sulphur dioxide	131.0 ppb
Nitrogen dioxide	104.6 ppb
Ozone	120.0 ppb
Carbon monoxide	8.6 ppb
Particulate matter (PM ₁₀)	50.0 µg/m ³
Lead	0.5 µg/m ³
Arsenic	6.0 ng/m ³
Cadmium	5.0 ng/m ³
Nickel	20.0 ng/m ³
Benzo(a)pyrene	1.0 ng/m ³
Benzene	1.66 ppb

2.3 Combined performance characteristic

- 2.3.1 MCERTS includes an overall measure of the performance of the CAM by combining together statistically all the relevant performance characteristics. This is known as a combined performance characteristic U_c . Only those performance characteristics that influence directly the uncertainty of the measurements produced by the CAM will be used to derive the combined performance, which shall be expressed as an expanded uncertainty with the level of confidence of 95%. The methodology used is based on the ISO Guide to the Expression of Uncertainty in Measurement (GUM)⁽⁶⁾ and on ISO Draft Standard 14956⁽⁷⁾. It is calculated by summing, in quadrature, the individual standard uncertainties u_i , determined for each relevant performance characteristic. The standard uncertainty u is determined by measurement and in the case of a normal distribution is usually taken as the standard deviation of the mean X_{bar} , as shown in Figure 1. The expanded uncertainty U is defined by multiplying the standard uncertainty u by a coverage factor to obtain a 95% confidence level. Therefore the 95% confidence interval for X lies within the limits defined by $\pm U$. The performance standards are defined in terms of measurement results and should be understood as being $\pm$ values. The numbers included in relevant tables are given as absolute values.

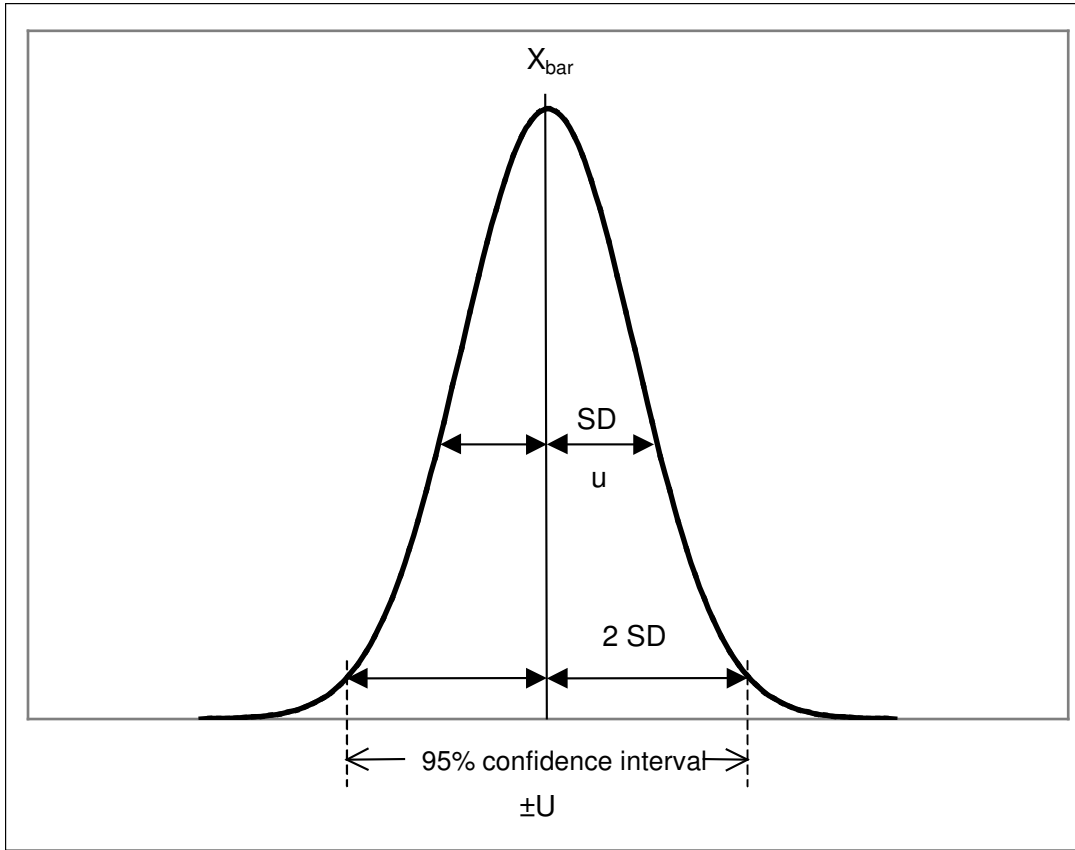


Figure 1

$$u_c = \sqrt{\sum_{i=1}^n u_i^2} \quad (2.1)$$

The combined performance characteristic U_c , expressed at the 95% confidence level, is derived from the combined standard uncertainty u_c by multiplying it by a coverage factor dependent on the total number of degrees of freedom.

- 2.3.2 The combined expanded uncertainties are compared against the MCERTS combined performance standards. The CAM must satisfy this requirement as well as meeting the performance standards specified for all individual performance characteristics. The combined performance standard will not be met if all individual performance characteristics contributing to the combined standard uncertainty are at their individual limits specified by MCERTS. Some of the individual performance characteristics have to be better than the corresponding performance standards if the combined performance standard requirement is to be met. On average, a CAM should have individual performance characteristics equal to half of the corresponding MCERTS performance standards, in order to meet this combined performance characteristic. Combined performance characteristic for CAMs monitoring CO, NO_x, SO₂, O₃ and particulates shall be calculated for an averaging period of one hour.

2.4 Operational conditions

2.4.1 CAM systems should be assessed under conditions which are representative of the most challenging intended application for which certification is sought. In rural and remote sites, very low concentration levels may be encountered, and CAMs should have low detection limits and minimal zero and span drift characteristics. In urban locations, including kerbside sites, CAMs are expected to monitor rapidly varying concentrations, over a wide dynamic range. Also these locations are likely to have higher levels of cross-interferent species which could have a greater effect on the CAM. CAMs are generally expected to measure under both of the above conditions and they will therefore be tested against the most challenging criteria.

2.4.2 Typical levels of pollutant concentrations measured in the UK are given in Table 2.4.

Table 2.4: Typical concentration values of the determinands covered in this document

Pollutant	Rural		Kerbside	
	Short term	Long term	Short term	Long term
Benzene	N/A	0.5 ppb annual average	N/A	2 ppb annual average
Carbon monoxide	200 ppb hourly average	100 ppb annual average	4 ppm hourly average	2 ppm annual average
Lead	N/A	0.01 µg/m ³ annual average	N/A	0.15 µg/m ³ annual average
Nitrogen dioxide	25 ppb hourly average	15 ppb annual average	55 ppb hourly average	40 ppb annual average
Ozone	70 ppb hourly average	35 ppb annual average	40 ppb hourly average	15 ppb annual average
Particulates (PM ₁₀)	35 µg/m ³ hourly average	25 µg/m ³ annual average	35 µg/m ³ annual average	25 µg/m ³ annual average
Sulphur dioxide	10 ppb hourly average	5 ppb annual average	30 ppb hourly average	15 ppb annual average

3 References

1. Procedures and General Requirements for the Testing of Continuous Ambient Air Quality Monitoring Systems. December 2002. Version 2.
2. European Air Quality Framework, Framework Directive, 96/62 EC, OJ L296, 21/11/1996.
3. European Air Quality Framework and First Daughter Directives, Framework Directive, 96/62 EC, Daughter Directive for SO₂, NO₂, PM₁₀ and Lead, 1999/30/EC, 22 April 1999.
4. Council Directive relating to limit values for benzene and carbon monoxide in ambient air, Directive 2000/69/EC, 16 November 2000.

5. The Air Quality Strategy for England, Scotland, Wales and Northern Ireland, DETR, January 2000.
6. Guide to the Expression of Uncertainty in Measurement, ISBN 92-67-10188-9, 1st ed, Geneva, Switzerland, ISO, 1993.
7. ISO 14956 Draft Standard, Evaluation of the Suitability of a Measurement Method by Comparison with a Stated Measurement Uncertainty.
8. Directive of the European Parliament and of the Council relating to arsenic, cadmium, mercury, nickel and polycyclic aromatic hydrocarbons in ambient air. PE-CON 3678/04, 28 October 2004.

4 Definitions of performance characteristics and other terms

Accuracy: The closeness of agreement between a single measured value of the determinand and the true value (or an accepted reference value).

Availability: Fraction of the total monitoring time for which data of acceptable quality have been collected (excluding servicing and calibrations).

Averaging effect: Measure of the uncertainty in CAM's results caused by short term fluctuations in the input concentrations within the time scale shorter than the averaging time T_a .

Averaging time: Period of time over which an arithmetic or time weighted average of concentrations is calculated [T_a – is the averaging period used by the CAM; T_{ra} is the required data averaging period, eg prescribed by legislation].

Combined performance characteristic: Expanded combined uncertainty calculated by summing the results of the relevant performance tests in accordance with the law of propagation of uncertainty, ie taking a square root of the sum of squares of the individual components.

Converter efficiency: Efficiency with which the internal converter unit of a NO_x analyser reduces NO_2 to NO .

Cross-sensitivity: Response of the CAM to determinands other than those that it is designed to measure.

Detection limit: The concentration value of determinand gas above which there is at least a 95 % degree of confidence that the measured value is different from zero.

Drift (short-term) – zero and span: Short-term zero and span drift is a measure of the drift in the output signal over the time period during which other performance characteristics are to be determined.

Dynamic retention efficiency: Ratio of the quantity of PAH extracted from sorbent and the quantity of PAH initially applied to the sorbent bed.

Effect of sample pressure change: Effect of sample gas pressure on the output signal of the CAM expressed in units of concentration per unit of pressure increment.

Effect of sample temperature change: Effect of sample temperature on the output signal of the CEM expressed in units of concentration per unit of temperature increment.

Expanded uncertainty: An interval about the measurement result that is expected to encompass a specific fraction of the distribution of values attributable to the determinand.

Field repeatability: Expanded uncertainty calculated from the standard deviation of differences between measurements made by two co-located identical analysers over the test period.

Flow-rate accuracy: The difference between the flow measured using a traceable standard and the mean flow rate.

Flow-rate stability: Assessment of statistical variations of flow in the sampling train during the test defined as a ratio of the standard deviation of the measured flow and the mean flow rate.

Linearity: Measure of fit of the instrument's response across its measurement range to a straight line using a number of samples of approximately equally distributed concentrations of a pollutant and a zero concentration.

Maintenance interval: Time in the operating environment in the field over which the CAM's zero and span drifts remain within specific limits.

Repeatability: Repeatability is a measure of the variation between successive measurements made maintaining the input span gas and all influence variables as constant as practicable.

Response time: A measure of the dynamic response of the CAM to a step change in the pollutant concentration [RT_{rise} is response time to a positive step change in pollutant concentration; RT_{fall} is response time to a negative step change in pollutant concentration].

Sampling efficiency: Ratio of the quantity of PAH extracted from the sorbent bed and the difference between the quantity initially applied to the sorbent bed and the quantity remaining on the sorbent bed.

Sample retention capacity: Percentage recovery determined individually for each of isotope-labelled PAHs with which sampling cartridges have been spiked prior to field test.

5 General instrument requirements

5.1 General requirements for CAMs submitted for testing

5.1.1 Manufacturers are required to submit the following:

- two identical, complete air quality monitoring systems (CAMs);
- calibration gases (where appropriate);
- for gaseous CAMs, all components required for the connection to a sampling manifold (but not the sampling manifold itself);
- all necessary components for operation under field conditions;
- for particulate monitors, all sampling components (including the sampling head);
- for systems (eg metals, PAHs) that require subsequent laboratory analyses, all sample collection, sample conditioning and sample handling systems.

5.1.2 The CAM submitted for testing shall have analogue or digital outputs that allow the provision of negative readings with respect to zero of at least 15% of the certification range, to ensure that the instrument performance tests carried out around the zero reading are valid (eg tests of zero drift, detection limit and cross-interference), and not truncated at the CAM's zero reading.

5.1.3 The determinand concentrations measured by a CAM are generally expressed either in density units (mass of determinand per unit volume of the ambient atmosphere) or in volume fraction units (volume of determinand per unit volume of the ambient atmosphere), although other units may also be used. The first of these units (density) is directly dependent on the temperature and pressure of the ambient atmosphere, while the second of these (volume fraction) may also have a weak dependence. Nationally traceable calibration gas mixtures, however, are generally defined in units of mole fraction, which are independent of gas temperature and pressure. Conversion factors between these different units may be calculated, but there may be uncertainties in some of the parameters in the conversion calculations. The CAM manufacturer or supplier shall inform the relevant MCERTS Certification Committee and the test house(s) as to which concentration units are being used, and the test house shall take these conversion calculations and any uncertainties into account when carrying out relevant tests. The test house and the MCERTS Certification Committee shall, where required, provide comments on the validity of any such algorithm employed by the CAM supplier or manufacturer, to ensure its proper subsequent use. It is therefore essential that the measurement units used by the CAM and any conversions to reference conditions applied (eg gas pressure and temperature) are clearly and unambiguously identified by the manufacturer. Results reported in units of mass per unit volume shall be expressed at standard temperature and pressure (temperature of 293 K, pressure of 101.3 kPa).

- 5.1.4 CAMs that have output readings sensitive to ambient air temperature and/or pressure shall be able to make corrections for changes in these parameters. These corrections may be carried out by using in-built pressure and temperature sensors or by using external sensors. The manufacturer or supplier of the CAM shall provide the test house with information as to whether any in-built temperature and pressure corrections are being applied. Where no internal corrections are applied, the manufacturer or supplier shall provide the test house with any algorithms that are required for the conversion of the CAM's readings to different ambient temperatures and pressures.
- 5.1.5 The CAM submitted for testing shall be in conformity with all applicable EC Directives. These include the Electro-magnetic Compatibility Directive 89/336/EEC, and its amendments 92/31/EEC and 93/68/EEC, and the Directive 72/23/EEC covering electrical equipment designed for use within certain voltage limits, and its amendment 93/68/EEC. Equipment within the scope of the Hazardous Atmospheres Directives falls outside the scope of this ambient MCERTS document. CAM manufacturers or suppliers shall supply declarations of conformity to all relevant Directives applicable to the equipment.

5.2 Response times

- 5.2.1 The response times RT_{rise} and RT_{fall} (as defined in Section 4) shall be determined as a part of the performance tests. A minimum performance requirement is that the CAMs shall have response times RT_{rise} and RT_{fall} less than 25% of the required averaging period. Table 5.1 lists examples of the averaging times used in the UK for collecting ambient air quality information. The performance standards for the response time take account of the residence times in a typical manifold, as these are not tested under the MCERTS Scheme. The minimum performance requirements for manifolds and/or sampling lines etc are given in Appendix 4.
- 5.2.2 CAMs shall have response times to positive and negative step changes in concentration that are within 10% (see Table 6.1) to ensure fast fluctuations of air quality concentrations are averaged correctly.
- 5.2.3 Some CAMs use adaptive filters for data smoothing. When testing the response times, the tests shall be performed with the filter enabled if this is the intended mode of operation. The response times measured shall be used to compare with the performance standards.
- 5.2.4 When testing NO_x CAMs, the response time shall be determined using both NO and NO_2 as the test gas. The response time for both gases will be compared with the performance standard.

5.3 Averaging times

- 5.3.1 Most of the performance requirements given in this document apply to results produced by CAMs that are averaged values of the pollutant concentration over a period defined as the averaging time T_a . In cases where the CAM internally produces averaged results and where the averaging time T_a is selectable, then it shall be selected

by the CAM manufacturer or supplier, in consultation with the MCERTS Certification Committee and the test house(s). In most cases the averaging times specified for air quality monitoring given in Table 5.1 should be used. The averaging times actually used will be stated on the MCERTS certificate.

Table 5.1 Examples of averaging times used for sampling and reporting in the UK

Pollutant	Typical averaging time (T_a^a) (also sampling time for metals, benzene and PAH CAMs)
Benzene	1 hour
Carbon monoxide	15 minutes
Lead	24 hour
Arsenic	24 hour
Cadmium	24 hour
Nickel	24 hour
Benzo(a)pyrene	24 hour
Nitrogen dioxide	15 minutes
Ozone	15 minutes
Particulates (PM ₁₀)	15 minutes
Sulphur dioxide	15 minutes

^adefined in Section 4.

5.4 Certification range

- 5.4.1 The CAM's manufacturer or supplier shall specify and agree with the MCERTS Certification Committee, for each pollutant concentration to be measured, a certification range of concentrations over which the CAM is to be tested for each determinand.
- 5.4.2 Each certification range shall be generally between zero and a maximum value of the pollutant concentration. These values shall be agreed by the MCERTS Certification Committee as being fit for the intended purpose.
- 5.4.3 Typical values of the certification range for CAMs are given in Table 5.2. These ranges are recommended unless the CAM manufacturer or supplier and the Certification Committee agree that there is a strong justification for selecting different ranges. Where no limit value is given, the compliance with the performance standards shall be verified at 80% of the certification range. If a manufacturer or supplier wishes to demonstrate performance over different ranges, additional testing will be required for each range.

Table 5.2 Typical certification ranges for CAMs

	Rural and remote sites (Category 1)		Urban background/centre, suburban, kerbside, roadside and industrial sites (Category 2)	
	Scale min.	Scale max.	Scale min.	Scale max.
NO	0.0 ppb	400 ppb	0.0 ppb	2000 ppb
NO ₂	0.0 ppb	400 ppb	0.0 ppb	2000 ppb
SO ₂	0.0 ppb	400 ppb	0.0 ppb	2000 ppb
CO	0.0 ppm	10 ppm	0.0 ppm	50 ppm
O ₃	0.0 ppb	500 ppb	0.0 ppb	500 ppb
PM ₁₀	0.0 µg/m ³	1000 µg/m ³	0.0 µg/m ³	1500 µg/m ³
PM _{2.5}	0.0 µg/m ³	1000 µg/m ³	0.0 µg/m ³	1500 µg/m ³

5.4.4 Where the CAM has user-selectable settings, range for example, these would be chosen by the CAM manufacturer or supplier and agreed with the Certification Committee in conjunction with the test house(s), to be appropriate for the certification range. In practice, the CAM range selected is likely to be similar to the certification range, although this is not essential. However, the CAM will be tested only over the certification range. The CAM settings, once chosen, will not be altered during the tests.

5.4.5 Auto-ranging CAMs will not have their ranges adjusted in any way by the test house(s) during the tests. The certificate will state that the performance tests have been carried out in auto-range mode. The certificate will show only the certification range tested and will not mention any internal range settings to which the CAM may have switched during the tests. If the CAM manufacturer or supplier wishes to certify the CAM at specific internal range settings, separate tests shall be performed, using agreed certification ranges. Where the output of an auto-ranging CAM is not converted to air quality units, or to a normalised scale, then an output must be available to flag which internal range setting applies to the output data signal.

5.4.6 The certificate will list all the ranges certified and the CAM settings used, and will state explicitly the performance characteristics tested.

5.5 Zero and span drift

5.5.1 During the field tests, the measurement of both the short-term and the long-term drift will be made by supplying certified zero and calibration gases externally to the sampling port of the CAM. Short-term drift tests will be performed every 12 hours for the first 14 days of the field tests, and the results obtained will be compared with the performance standards.

5.5.2 In addition, the CAM will have associated with it, where possible, a methodology using either an automated or a manual method to measure the zero drift and span drift of the complete system. The CAM supplier or manufacturer shall provide a

description of the methodology used to determine these drifts. This will be assessed during the field tests.

- 5.5.3 In CAMs where the daily zero and span checks are used to automatically change the zero response and calibration factor of the CAM, this auto-scaling function will be disabled during certification.

5.6 NO_x analysers

- 5.6.1 NO₂ is not directly measured by most current NO_x CAMs, but instead is obtained by subtracting the measured NO concentration from the measured NO_x concentration.
- 5.6.2 During the averaging tests, the NO_x and NO channels will be monitored to check if the CAM is reporting negative NO₂ concentrations. If negative values are observed the CAM shall be deemed unsuitable for Category 2 applications.
- 5.6.3 When the NO₂ concentrations approach the limit value in an urban background/centre scenario, the accompanying concentrations of NO are typically 5 - 6 times the NO₂ concentrations. Therefore testing of the CAM's performance characteristics with NO test gas shall be done at a NO concentration of about 600 ppb, equal to approximately six times the NO₂ limit value. This applies to the following performance characteristics: the averaging effect, linearity, cross-sensitivity, environmental conditions, effect of sample temperature and pressure.

5.7 Sampling line, sampling manifold and particulate filter

- 5.7.1 Sampling manifolds will not be tested under the MCERTS Scheme. However, Appendix 4 provides recommended minimum requirements for the design of manifolds and sample lines.
- 5.7.2 For gaseous CAMs, a particulate filter is generally installed to precondition the sample before it enters the analytical unit of the CAM. This particulate filter shall retain all particulate matter likely to alter the performance of the CAM analytical unit or contaminate its internal components. The filter housing shall be so constructed that it does not retain or convert the determinand(s) of interest. Tests of the CAM will be carried out with this particulate filter included.

5.8 Cross-sensitivity to interfering substances

- 5.8.1 The CAM's response to interferent gases shall be tested. The interferent gas concentrations shall be set at levels appropriate for different measurement category locations.
- 5.8.2 Cross-sensitivity tests shall be performed using binary mixtures of each interferent gas with the determinand gas at the limit value given in Table 2.3 and using a mixture of all interferent gases with the determinand gas at the limit value. This tests the assumption of additivity of the CAM's cross-sensitivity response.

- 5.8.3 Samples containing the determinand gas alone at the limit concentration shall be introduced to the CAM between the interference tests to provide a reference response.

5.9 Test of linear fit

- 5.9.1 This performance standard is to be specified as the percentage of the measured value. Linearity shall be assessed at six points approximately equally spaced across the measurement range, with one point at zero concentration. The differences between the best-fit line and an average at each point expressed as percentages of the measured value will be compared with the performance standard.

- 5.9.2 When deciding on the NO concentrations to be used for the test of linear fit for NO_x CAMs, the lowest non-zero test concentration shall be at or below the NO₂ limit value concentration.

6. Determinand specific requirements

6.1 Introduction

- 6.1.1 This section defines the determinand specific requirements for CAMs.
- 6.1.2 The performance standards are expressed in terms of measurement results from which standard uncertainties can be derived, in keeping with international procedures.

6.2 Performance standards for CAMs measuring SO₂, NO_x, O₃ and CO

- 6.2.1 CAMs used for monitoring SO₂, NO_x, O₃ and CO draw in a sample of ambient air, usually from a manifold, into an analytical system where the concentration of the relevant pollutant is measured. For NO_x monitoring systems, however, a catalytic converter is used to reduce NO₂ to NO, and two measurements of NO are made, with and without the converter, to allow the NO₂ concentration to be determined by difference.

The MCERTS performance standards are given in Table 6.1.

Table 6.1 Analyser performance characteristics and assessment criteria

NOTE nmol/mol = ppb

Item	Performance characteristic	Lab test	Field test	Performance criterion			
				SO ₂	NO _x	CO	O ₃
1	Repeatability at zero	x		≤ 1.0 nmol/mol	≤ 1.0 nmol/mol	≤ 1.0 μmol/mol	≤ 1.0 nmol/mol
2	Repeatability at hourly limit value	x		≤ 3.0 nmol/mol	≤ 3.0 nmol/mol	≤ 3.0 μmol/mol	≤ 3.0 nmol/mol
3	Residual lack of fit at zero	x		≤ 5.0 nmol/mol	≤ 5.0 nmol/mol	≤ 0.2 μmol/mol	≤ 5.0 nmol/mol
4	Lack of fit (largest residual from the linear regression line)	x		≤ 4 % of the measured value ^b	≤ 4 % of the measured value ^b	≤ 4% of the measured value ^b	≤ 4 % of the measured value ^b
5	Sensitivity coefficient to sample gas pressure	x		≤ 3.0 nmol/mol/kPa	≤ 8.0 nmol/mol/kPa ^a	≤ 0.7 μmol/mol/kPa ^a	≤ 2.0 nmol/mol/kPa
6	Sensitivity coefficient to sample gas temperature	x		≤ 1.0 nmol/mol/K	≤ 3.0 nmol/mol/K	≤ 0.3 μmol/mol/K	≤ 1.0 nmol/mol/K
7	Sensitivity coefficient of surrounding air temperature	x		≤ 1.0 nmol/mol/K	≤ 3.0 nmol/mol/K	≤ 0.3 μmol/mol/K	≤ 1.0 nmol/mol/K
8	Sensitivity coefficient of electrical supply voltage	x		≤ 0.3 nmol/mol/V	≤ 0.3 nmol/mol/V	≤ 0.3 μmol/mol/V	≤ 0.3 nmol/mol/V
9	Converter efficiency	x		-----	> 98%	-----	-----
10	Interference by H ₂ O ^c	x		≤ 10 nmol/mol at concentration of 19 mmol/mol ^d	≤ 5 nmol/mol at concentration of 19 mmol/mol ^d	≤ 1 μmol/mol at concentration of 19 mmol/mol ^d	≤ 10 nmol/mol at concentration of 19 mmol/mol ^d
11	Interference by H ₂ S ^c	x		≤ 5 nmol/mol at concentration of 200 nmol/mol	-----	-----	-----
12	Interference by NH ₃ ^c	x		≤ 5 nmol/mol at concentration of 200 nmol/mol	≤ 5 nmol/mol at concentration of 200 nmol/mol	-----	-----
13	Interference by NO ^c	x		≤ 5 nmol/mol at concentration of 500 nmol/mol	-----	≤ 0.5 μmol/mol at concentration of 1 μmol/mol	-----
14	Interference by NO ₂ ^c	x		≤ 5 nmol/mol at concentration of 200 nmol/mol	-----	-----	-----
15	Interference by m-xylene ^c	x		≤ 10 nmol/mol at concentration of 1 μmol/mol	-----	-----	≤ 5 nmol/mol at concentration of 0.5 μmol/mol
16	Interference by toluene ^c	x		-----	-----	-----	≤ 5 nmol/mol at concentration of 0.5 μmol/mol
17	Interference by CO ₂ ^c	x		-----	≤ 5 nmol/mol at concentration of 500 μmol/mol	≤ 0.5 μmol/mol at concentration 500 μmol/mol	-----

Table 6.1 Analyser performance characteristics and assessment criteria (continued)

Item	Performance characteristic	Lab test	Field test	Performance criterion			
				SO ₂	NO _x	CO	O ₃
18	Interference by N ₂ O ^c	x		-----	----- --	≤ 0.5 µmol/mol at concentration of 50 nmol/mol	-----
19	Interference by ozone ^c	x		-----	≤ 2 nmol/mol at concentration of 200 nmol/mol	-----	-----
20	Averaging effect	x		≤ 7% of the measured value	≤ 7% of the measured value	≤ 7% of the measured value	≤ 7% of the measured value
21	Reproducibility under field conditions ^f		x	≤ 5% of the average over a three months period	≤ 5% of the average over a three months period	≤ 5% of the average over a three months period	≤ 5% of the average over a three months period
22	Long term drift at zero		x	≤ 5 nmol/mol	≤ 5 nmol/mol	≤ 0.5 µmol/mol	≤ 5 nmol/mol
23	Long term drift at span level ^a		x	≤ 5% of maximum of certification range	≤ 5% of maximum of certification range	≤ 5% of maximum of certification range	≤ 5% of maximum of certification range
24	Short term drift at zero	x		≤ 2 nmol/mol over 12 h	≤ 2 nmol/mol over 12h	≤ 0.10 µmol/mol over 12 h	≤ 2 nmol/mol over 12 h
25	Short term drift at span level ^a	x		≤ 6 nmol/mol over 12 h	≤ 6 nmol/mol over 12 h	≤ 0.60 µmol/mol over 12 h	≤ 6 nmol/mol over 12 h
26	Response time (rise)	x		180 s	180 s	180 s	180 s
27	Response time (fall)	x		180 s	180 s	180 s	180 s
28	Difference between rise and fall time	x		≤ 10% or 10 seconds	≤ 10% or 10 seconds	≤ 10% or 10 seconds	≤ 10% or 10 seconds
29	Difference between sampling and calibration port	x		≤ 1%	≤ 1%	≤ 1%	≤ 1%
30	Difference in NO ₂ due to residence time in analyser	x		-----	≤ 4 nmol/mol	-----	-----
31	Period of unattended operation		x	3 months but not less than two weeks	3 months but not less than two weeks	3 months but not less than two weeks	3 months but not less than two weeks
32	Availability of analyser		x	> 90 %	> 90 %	> 90 %	> 90 %
33	Total expanded uncertainty	x	x	15%	15%	15%	15%

^a span level is 70-80% of the certification range

^b and 2 nmol/mol at zero (0.2 µmol/mol for CO analysers)

^c performance criterion is set both at zero and span level

^d H₂O concentration of 19 mmol/mol equals 80% RH at 293 K and 101,3 kPa

^e if relevant

^f if the average concentrations at the monitoring site are less than 0.3 of the LV then 0.3 of the LV shall be used instead of average

6.3 Performance standards for CAMS measuring particulate matter (PM₁₀ and PM_{2.5})

6.3.1 Particulate monitoring CAMs collect particulate material over a defined averaging time (T_a) and either:

- dynamically determine the incremental mass increase; by direct mass measurements, eg by using a tapered element oscillating micro-balance (TEOM), or by a surrogate technique eg a β -attenuation gauge; or
- retain the collected material for subsequent analysis (weighing); or
- apply other measurement principle such as for example light scattering.

The sampling system is an inherent component of gravimetric particulate monitors, as they require size-selective sampling heads to collect particulate material preferentially according to its aerodynamic size. Two categories of particulate material are currently monitored, PM₁₀ and PM_{2.5}. These refer to the aerodynamic size of ambient particulates collected with sampling heads for which a 50% cut-off is achieved.

MCERTS performance standards for particulate monitors are given in Table 6.3

6.4 Field testing of instruments measuring particulate matter (PM₁₀ and PM_{2.5})

6.4.1 General principles

Field tests are performed with the instrument operated side by side with the standard reference method. The purpose of the measurements is to assess the uncertainty between the instruments through the use of two instruments working in parallel and the comparability of the instruments with the standard reference method. It is acceptable to use the results of existing studies provided that the requirements of the relevant sampling standards have been met.

In order to assess proper functioning of the standard reference method two reference samplers or instruments may be used. The mean squared difference of the results produced by both reference samplers provides an estimate of the random uncertainty of the reference method.

6.4.2 Experimental conditions

The test sites selected for the field tests should be representative of the conditions under which the instruments are likely to operate. Occurrences of possible episodes of high concentrations are desirable. A minimum of four comparisons should be carried out with the emphasis on the following variables:

- Composition of the PM fraction, preferably including high and low fractions of semi-volatile fractions to be able to cover maximum impact of potential losses;

- High and low air temperature and humidity to cover any conditioning losses of semi-volatiles;
- Large variations of wind speed to cover the impact on sampling inlet performance.

During the tests the following information should be collected and recorded:

- Calibration procedures, equipment and intervals;
- Results of the quality assurance checks;
- Temperature and pressure of the sampled air;
- Humidity and other relevant conditions;
- Events likely to affect the measurement results.

A minimum of 40 measurement results, each averaged over a period of 24 hours constitutes a comparison run. Of the full data set at least 20% of the results shall be greater than 50% of the limit value specified in the relevant Daughter Directive.

6.4.3 Calculation of performance characteristics

Uncertainty between the instruments is calculated using the methodology and formula given below.

Results of the instrument under tests are averaged over 24 hour measurement period i to produce a set of values y_i . The uncertainty between the instruments u_{bs} is calculated from the differences of all 24 hour results operated in parallel:

$$u_{bs}^2 = \frac{\sum_{i=1}^n (y_{i,1} - y_{i,2})^2}{2n}$$

where,

$y_{i,1}$ and $y_{i,2}$ are the results of parallel measurement for a single 24 hour period i
 n = number of 24 hour measurement results

The uncertainty between the instruments should be determined:

- For each of the comparisons separately;
- For all the results together;
- For two data sets obtained by splitting the whole data set according to PM concentrations; greater than or equal to upper assessment threshold or below the upper assessment threshold specified in the relevant Daughter Directive.

Table 6.2 Performance standards for CAMs measuring PM₁₀ and PM_{2.5} (field tests)

Performance characteristic	Performance standard
Between sampler uncertainty for all the comparison runs separately	< 3 µg/m ³
Between sampler uncertainty for all results together	< 3 µg/m ³
For two data sets split into values below and above 50% of the upper assessment threshold	< 3 µg/m ³

Uncertainty between the instruments greater than 3 µg/m² is an indication of unsatisfactory performance of the systems in any of the above scenarios and equivalence with the standard reference method can not be declared.

6.4.4 Comparison with the standard reference method

The performance of the standard reference method can be evaluated by checking the uncertainty between the reference instruments. This uncertainty is satisfactory if the result is $\leq 2 \mu\text{g}/\text{m}^3$.

For the evaluation of uncertainty due to the lack of comparability between the instrument under tests and the standard reference method it is assumed that the relationship between both measurements can be described by a linear equation in the form:

$$y_i = a + bx_i$$

The equation coefficients are calculated using a regression technique involving symmetrical treatment of both variables. A commonly applied computational technique is that of orthogonal regression. The regression equation is calculated for:

- Each of the comparison runs separately;
- All the results combined;
- For a data set containing concentrations greater than or equal to 50% of the upper assessment threshold specified in the relevant Daughter Directive.

When a value of b is outside the range of 0.9 to 1.1 a correction factor is applied to the dataset as described in the report of the EC Working Group on Demonstration of Equivalence of Ambient Air Monitoring Methods (see Appendix 1).

Further evaluation procedure involves calculation of the uncertainty in the instrument results compared with the standard reference method u_{c-s} using a general equation describing u_{c-s} as a function of PM concentrations x_i . The functional relationship between u_{c-s} and x_i is given by:

$$u_{c-s}^2(y_i) = \frac{RSS}{(n-2)} - u^2(x_i) + [a + (b-1)x_i]^2$$

Where,

RSS - sum of relative residuals resulting from orthogonal regression

$U(x_i)$ - random uncertainty of the reference method, for example the value of u_{bs}

The sum of relative residuals RSS is calculated using the equation:

$$RSS = \sum_{i=1}^n (y_i - a - bx_i)^2$$

The uncertainty U_{c-s} is calculated for each of the comparison runs separately and for all the results combined.

6.4.5 Calculation of the combined uncertainty of measurement

For all the data sets the combined relative uncertainty $w_{c,CM}$ of the instrument is calculated by combining the contributions using the formula:

$$w_{c,CM}^2(y_i) = \frac{u_{c-s}^2(y_i)}{y_i^2}$$

For each of the datasets the uncertainty at the daily limit value w_{CM} is calculated taking the concentration at the limit value as y_i . The expanded relative uncertainty is calculated for each of the data sets by multiplying $w_{c,CM}$ by a coverage factor k corresponding to the appropriate number of the degrees of freedom.

$$W_{CM} = k \cdot w_{CM}$$

Because the number of the experimental results is large the coverage factor of 2 is normally used.

6.4.6 Evaluation of the field results

The highest resulting uncertainty estimate W_{CM} is compared with the expanded relative uncertainty based on the data quality objectives W_{dgo} . The criteria for acceptance or rejection are as follows:

$W_{CM} \leq W_{dgo}$ the instrument is accepted as equivalent to the standard reference method.

$W_{CM} > W_{dgo}$ the instrument is not accepted as equivalent to the standard reference method.

6.5 Performance standards for metals-measuring CAMs

- 6.5.1 Airborne particles containing heavy metals (lead, cadmium, arsenic, nickel and mercury) are collected by drawing a measured volume of air through a filter mounted in a sampler designed to collect the PM₁₀ (or PM_{2.5}) fraction of the total suspended particulate matter in ambient air. The performance standards cover only the collection of particulate matter, and do not include its subsequent analysis, which should be carried out by analytical laboratories appropriately accredited to ISO 17025. It is assumed that the collection of particulate matter with the required measurement uncertainty constitutes a satisfactory analogue for the collection of the metals content.
- 6.5.2 Metals-measuring CAMs shall comply with the performance standards specified for particulates. Table 6.3 gives the specific performance standard for metals-measuring CAMs.

Table 6.3 Performance standards for metals-measuring CAMs

Performance characteristic	Metal	Performance standard
Total expanded uncertainty ^b	Arsenic Cadmium Nickel	40% LV ^a
Total expanded uncertainty ^b	Lead	25% LV ^a

^aLV = Limit Value in accordance with relevant Directive (if applicable) or set by the Certification Committee as a default value.

^bThe uncertainty of laboratory analysis is included.

- 6.5.3 The required total uncertainty for the measurement of some metals is set by the EU Directive at 40% for arsenic, cadmium and nickel and at 25% for lead, and combines the uncertainty of sample collection and the uncertainty of analysis. MCERTS does not determine any uncertainties due to laboratory analysis which should be specified by the accredited laboratory carrying out the analysis and included in the calculations. Uncertainties in laboratory analysis of particulate metals collected on filters should generally be better than 10% (95% confidence level). The combined performance characteristic given in Table 6.4 contains an allowance for this uncertainty of analysis.
- 6.5.4 In the case of metals-measuring CAMs, the choice of filter medium is particularly important. The following requirements on the background content of metals within the filters used are therefore provided as guidance.
- 6.5.5 Only those filter media that retain $\geq 0.3 \mu\text{m}$ diameter particulates with an efficiency of $>99\%$ should be employed. All filter materials should have a low metal background

content, not worse than that given for PTFE in Table 6.4. In addition, care must be exercised in the handling and conditioning of filters.

Table 6.4 Maximum blank levels of selected heavy metals on PTFE filters

Metal	Blank levels of metal on filter (ng/cm²)	Background level^a on 47 mm filter (ng)
Pb	2.0	35
As	0.8	14
Ni	0.3	6
Cd	0.7	10
Hg	0.4	7

^aBackground levels derived from one-week samples and a sample flow of 16.7 litre/min.

- 6.5.6 Filters made from other media (eg cellulose nitrate, quartz fibre or glass) are not recommended due to their relatively high heavy-metal content.

6.6 Performance standards for CAMs measuring polycyclic aromatic hydrocarbons (PAHs)

- 6.6.1 Limit value is currently specified for PAHs (benzo(a)pyrene) in the new Daughter Directive relating to arsenic, cadmium, mercury, nickel and polycyclic aromatic hydrocarbons in ambient air.
- 6.6.2 The manufacturer or supplier of the CAM shall agree with the MCERTS Certification Committee the specific PAH species and the concentration range over which product certification is sought, and the species and range over which MCERTS certification is granted will be reported in the certificate. A list of possible PAHs that may be tested is given in Appendix 2.
- 6.6.3 The relevant Directive specifies overall measurement uncertainty requirements in Europe for the reporting of PAHs at the level of 50% including sampling and analysis. MCERTS does not specify the uncertainty for the laboratory analysis which should be assessed by the accredited laboratory carrying out the tests and included in the overall calculations.
- 6.6.4 The MCERTS performance standards for PAH-monitoring CAMs are given in Table 6.5 and 6.6. They include the efficiency tests on filter materials for the collection and retention of PAHs but do not include the contribution of analysis.
- 6.6.5 A mixture of two PAHs out of the set for which the certification is sought will be used for laboratory testing. The volatility of the test PAHs used will cover the whole volatility range of the PAHs for which certification is sought.

- 6.6.6 As sampling is carried out over at least a 24-hour period, measurements of response time and short-term drift (12 hour) are not required.

Table 6.5 Performance standards for PAHs (laboratory tests)

Performance characteristic	Performance standard
Uncertainty due to sampling efficiency	$\leq 25\% \text{ LV}^a$
Uncertainty due to retention efficiency	$\leq 25\% \text{ LV}^a$
Detection limit for 24 hour sampling time	$\leq 18 \text{ ng}$ (equivalent to 0.05 ng/m^3 drawn at 15 m^3 per hour for 24 hours)
Combined performance ^b	$\leq 25\% \text{ LV}^2$

^aThe uncertainty of the laboratory analysis of PAHs is not included.

^bExpressed as percentage of LV (if it exists) or a default value set by the Certification Committee.

Table 6.6 Performance standards for PAHs (field tests)

Performance characteristic	Performance standard
Uncertainty due to field repeatability	$\leq 10\% \text{ LV}^a$
Uncertainty due to sample retention capacity	$\leq 25\% \text{ LV}^a$
Uncertainty due to flow-rate accuracy	$\leq 10\% \text{ LV}^a$
Uncertainty due to flow-rate stability	$\leq 10\% \text{ LV}^a$

^aLV Limit Value set in accordance with relevant Directive (if applicable) or set by the certification committee as a default value

6.7 Performance standards for benzene-monitoring systems

- 6.7.1 These performance standards apply to cyclic automated CAMs, which draw in a known volume of air through an adsorbent tube for a fixed time, and then analyse the amount of benzene trapped. The sampling period is by definition T_a and consists of the sampling time and the analysis time. The CAM's performance when measuring rapidly varying concentrations shall also be tested. Tests for long-term sampling CAMs, which do not analyse samples on-line, are not covered in this document.

- 6.7.2 The time coverage factor of the CAM, related to its cyclic mode of operation and the fraction of time that ambient air is sampled, shall be calculated and stated on the MCERTS certificate.
- 6.7.3 As these CAMs use an absorbent to collect the sample before analysis, the effect of desorption inefficiencies shall be assessed by measuring the amount of benzene carried over into an additional adsorption stage.
- 6.7.4 The instrument performance standards for benzene-monitoring CAMs are given in Tables 6.7 and 6.8.

**Table 6.7 Performance standards for benzene
(laboratory tests)**

Performance characteristic	Performance standard
<i>Included in the uncertainty budget</i>	
Carry-over	≤ 0.12 ppb
Repeatability at span	≤ 0.12 ppb
Averaging effect	≤ 0.20 ppb
Accuracy	≤ 0.20 ppb
Linearity	≤ 5% of measured value
Cross-interference	≤ 0.2 ppb
Effect of sample pressure per 1 kPa	≤ 0.01 ppb
Zero drift due to ambient temperature per 10 K	≤ 0.06 ppb
Dependence of line voltage per 10 V	≤ 0.04 ppb
<i>Not included in the uncertainty budget</i>	
Repeatability at zero	≤ 0.005 ppb
Short-term zero drift	≤ 0.06 ppb
Short-term span drift	≤ 0.06 ppb
Combined expanded uncertainty ^a	≤ 25% of limit value

^aExpressed as an expanded uncertainty of the limit value concentration, with a 95% confidence level.

Table 6.8 Performance standards for benzene (field tests)

Performance characteristic	Performance standard
Long-term zero drift	≤ 0.26 ppb
Long-term span drift	≤ 0.26 ppb
Repeatability in the field	≤ 0.12 ppb
Maintenance interval	≥ 14 days
Data capture	≤ 90%

7 Status of this document

- 7.1 Version 4 of the MCERTS Performance Standards for Continuous Ambient Air Quality Monitoring Systems replaces version 3, which is withdrawn and should now be discarded. Version 4 may be subject to review and amendment following publication. For the latest version go to the environment Agency's website at :

www.mcerts.net

Appendix 1: Main standards and CEN activities underpinning the MCERTS performance standards

A1.1 Main standards reviewed

Determinand	International standard used as basis for performance standards
General	ISO 6879 (1995) Air quality – Performance characteristics and related concepts for air quality measurements ISO 9169 (1994) Air quality –Determination of performance characteristics of measurement methods. Also Amendment 1 (CD) (1998) BS EN 61207-1 (IEC 1207-1) 1994 Expression of performance of gas analyzers
SO ₂	BS EN 14212:2005 Ambient air quality - Standard method for the measurement of the concentration of sulphur dioxide by ultraviolet fluorescence
CO	ISO/DIS 4224 Ambient air- Determination of carbon monoxide - Non-dispersive infrared spectrometric method BS EN 14626:2005 Ambient air quality– Standard method for the measurement of carbon monoxide concentrations in ambient air by non-dispersive infrared spectroscopy
NO _x	BS EN 14211 Ambient air quality- Standard method for the measurement of the concentration of nitrogen dioxide and nitrogen monoxide by chemiluminescence ISO 7996:1985 Ambient air – Determination of the mass concentration of nitrogen oxides – Chemiluminescence method
O ₃	BS EN 14625 Ambient air quality – Standard method for the measurement of ozone by ultraviolet photometry ISO 10313:1993 Ambient air – Determination of the mass concentration of ozone – Chemiluminescence method
Particulates	EN 12341 Air quality – Determination of the PM₁₀ fraction of suspended particulate matter – Reference method and field test procedure to demonstrate reference equivalence of measurement methods PrEN 14907 Ambient air quality - Reference gravimetric measurement method for the determination of the PM 2.5 mass fraction of suspended particulate matter ISO/DIS 10473 Ambient air – Measurement of the mass of particulate matter on a filter medium – beta-ray absorption method
Lead	ISO 9855:1993 Ambient air – Determination of the particulate lead content of aerosols collected on filters – atomic absorption spectrometric method
PAHs	ISO/DIS 12884 Air quality – Determination of gas- and particle-phase polycyclic aromatic hydrocarbons in ambient air - Collection on sorbent-backed filters with gas chromatographic/mass spectrometric analysis

Benzene	BS EN 1076:1997 Workplace atmospheres. Pumped sorbent tubes for the determination of gases and vapours. Requirements and test methods VDI 3482 Part 5 Measurement of gaseous emissions - Gas chromatographic determination of aromatic hydrocarbons - sampling by enrichment on activated carbon -desorption with solvent
Uncertainty	Draft EN ISO 20988 Air Quality - Guidelines for estimating measurement uncertainty
Equivalence	EC Working Group on Guidance for Demonstration of Equivalence Demonstration of Equivalence of Ambient Air Monitoring Methods
Metals	PrEN Ambient Air Quality - Standard method for the measurement of Pb, Cd, As and Ni in the PM 10 fraction of suspended particulate matter

A1.2 CEN technical committee and working groups

CEN technical committee and associated working groups	Title
CEN TC 264	Air Quality
WG 12	Reference Methods for SO ₂ , NO ₂ , O ₃ and CO in Ambient Air
WG 13	Reference Methods for Benzene in Ambient Air
WG 14	Reference Methods for Pb, Cd, As and Ni in Ambient Air
WG 15	Reference Methods for PM _{2.5} Measurements in Ambient Air

Appendix 2: List of PAHs to be considered

This list of PAHs is based on those monitored by the DETR's Toxic Organic Micro Pollutants (TOMPs) Network.

Acenaphthene
Acenaphthylene
Anthracene
Benz[*a*]anthracene
Benzo[*a*]pyrene
Benzo[*b*]fluoranthene
Benzo[*k*]fluoranthene
Chrysene
Coronene
Dibenz[*ac*]anthracene
Dibenz[*ah*]anthracene
Fluoranthene
Fluorene
Indeno[123,*cd*]pyrene
Methylphenanthrene
Phenanthrene
Pyrene

Appendix 3: Standard environmental operating conditions

Several existing IEC standards depend upon the first edition of IEC Publication 359, and call for the manufacturer to specify particular values and ranges of influence quantities, according to three degrees of severity known as Usage Groups I, II and III, as follows:

- I For indoor use and under conditions that are normally found in laboratories and factories and where equipment would be handled carefully.
- II For use in environments having protection from full extremes of environment and under conditions of handling between those of Groups I and III.
- III For outdoor use and in areas where the equipment may be subjected to rough handling.

For testing to the proposed MCERTS performance standards, it will be assumed that CAM are operated and tested to Group I conditions.

A3.1 Climatic conditions

A3.1.1 Ambient temperature

Rated ranges of use: +5°C to +40°C

Storage and transport temperature: -40°C to +70°C

A3.1.2 Relative humidity of the air

Because extreme values of both temperature and humidity are not likely to occur simultaneously, the manufacturer may specify the time limit over which these may be applied and shall specify the limitations of the combination, if any, for continuous operation.

Rated ranges of use: 20% to 80% excluding condensation

A3.1.3 Barometric pressure (altitude)

Rated ranges of use: 70.0 kPa to 106.0 kPa
(up to 2200 m altitude)
(525 mmHg to 800 mmHg)

A3.1.4 Heating effect due to solar radiation

Rated ranges of use: no direct irradiation

A3.1.5 Velocity of the ambient air

Rated ranges of use: 0 m/s to 0.5 m/s

A3.1.6 Sand and dust content of the ambient air

Rated ranges of use: negligible content

A3.1.7 Salt content of the air

Rated ranges of use: negligible content

A3.1.8 Contaminating gas or vapour content of the air

Rated ranges of use: specified by the manufacturer

A3.1.9 Liquid water content of the air

Rated ranges of use: negligible content

A3.2 Mechanical conditions

A3.2.1 Operating position

Rated ranges of use: recommended position $\pm 30^\circ$

A3.2.2 Ventilation

Rated ranges of use: negligibly obstructed

A3.2.3 Vibration

Rated ranges of use: negligible value

A3.3 Mains supply conditions

A3.3.1 Mains supply voltage (considering a distorted waveform)

	DC and AC (rms) (% of rated voltage)	AC peak (% of rated voltage)
Rated ranges of use	$\pm 10\%$	$\pm 12\%$
Limit range of operation (called “limiting conditions” in this standard)	Equal to the rated range of use, unless otherwise stated	

A3.3.2 Mains supply frequency

Rated ranges of use: rated value $\pm 5\%$

A3.3.3 Distortion of AC mains supply

The distortion is determined by a factor b in such a way that the waveform is inside an envelope formed by:

$$y_1 = (1 + b) A \sin wt$$

and

$$y_2 = (1 - b) A \sin wt$$

Rated ranges of use: $b = 0.05$

The values of b are valid when the apparatus is connected to the supply mains.

Notes: The above formulae may be applied over a half cycle or a full cycle depending on whether the zero crossings are equally spaced or not.

A3.3.4 Other distortions

Pulses and spikes in an AC mains supply are provisionally covered by mains supply frequency.

A3.3.5 Ripples of DC supply

Rated ranges in use:	0.5% of the supply voltage
Limit range of operation:	5.0% of the supply voltage

The values given are peak-to-peak values of the ripple voltage expressed as a percentage of the average DC supply voltage.

Appendix 4: Recommendations for sampling lines and manifolds

- A4.1 The sample intake should be constructed in such a way that suction of rain water into the sample line (or CAM system) is prevented. The material of the sample intake as well as the sample line (or CAM system) can influence the composition of the gas in the sampling line. Materials such as polytetrafluoroethylene (PTFE), perfluoro-ethylene-propylene (FEP), glass or stainless steel may be used. Copper or copper-based alloys should not be used. If condensation due to high ambient temperatures and/or humidity occurs, the sample lines may be moderately heated.
- A4.2 In the case of NO_x measuring CAMs, in order to avoid disturbance of the nitrogen monoxide-nitrogen-dioxide-ozone equilibrium in the sample line (or system), the residence time in the sample system (or line) from the sampling inlet to the monitor should be such that the increase of the nitrogen dioxide content in the sample is less than 2%. The influence of the pressure drop along the complete sampling system, including any filter, should be such that it causes an apparent concentration change of less than 1% of the signal output of the analyser.
- A4.3 The increase of the sampled nitrogen dioxide is due to the effect of the reaction of ambient ozone with nitric oxide in the sampling line. The influence of the residence time on the increase of nitrogen dioxide in the sampling line can be estimated by the following formula:

$$[O_3]_0 = \frac{b \cdot [O_3]_t}{[O_3]_t - [NO]_t \cdot e^{(b \cdot k \cdot t)}} \quad (A4.1)$$

where:

- $[O_3]_0$ is the ozone concentration at the sample manifold inlet;
 $[O_3]_t$ is the ozone concentration after t seconds of residence time in the sampling line;
 $[NO]_t$ is the nitric oxide concentration after t seconds of residence time in the sampling line;
 b is the concentration difference between $[O_3]_t$ and $[NO]_t$: $b = [O_3]_t - [NO]_t$ with $b \neq 0$;
 k is the rate constant for the reaction of O₃ with NO: $k = 4.43 \times 10^{-4} \text{ ppb}^{-1} \text{ s}^{-1}$ at 25°C;
 t is the residence time in seconds,

and

$$[O_3]_0 - [O_3]_t = [NO]_0 - [NO]_t$$

where

- $[NO]_0$ is the NO concentration at the sample manifold inlet

Example

Assuming a residence time of 2 seconds in the manifold and the following concentrations measured after the sampling manifold:

$$[O_3]_t = 22 \text{ ppb}$$

$$[NO]_t = 100 \text{ ppb}$$

Equation (A4.1) will give an ozone concentration at the sample inlet, $[O_3]_0$, of 24.1 ppb and therefore a NO concentration at the sample inlet of 97.9 ppb. This would result in an increase of 2.1 ppb of NO_2 due to the reaction of NO and O_3 in the sampling manifold and a loss of 2.1 ppb of NO.